Loss Given Default (LGD) Model Report

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Primary Model: XGBoost Regressor (LGD Severity)
Benchmark Model: Random Forest Regressor

1. Executive Summary & Objective

This report validates the Loss Given Default (LGD) model developed for the retail lending portfolio. In credit risk analytics, LGD represents the percentage of exposure lost if a borrower defaults. The target is defined as (loan_amnt - recoveries) / loan_amnt, capped within [0.0, 1.0]. The model is trained purely on the cohort of defaults using application-time features (no leakage from post-origination behaviors).

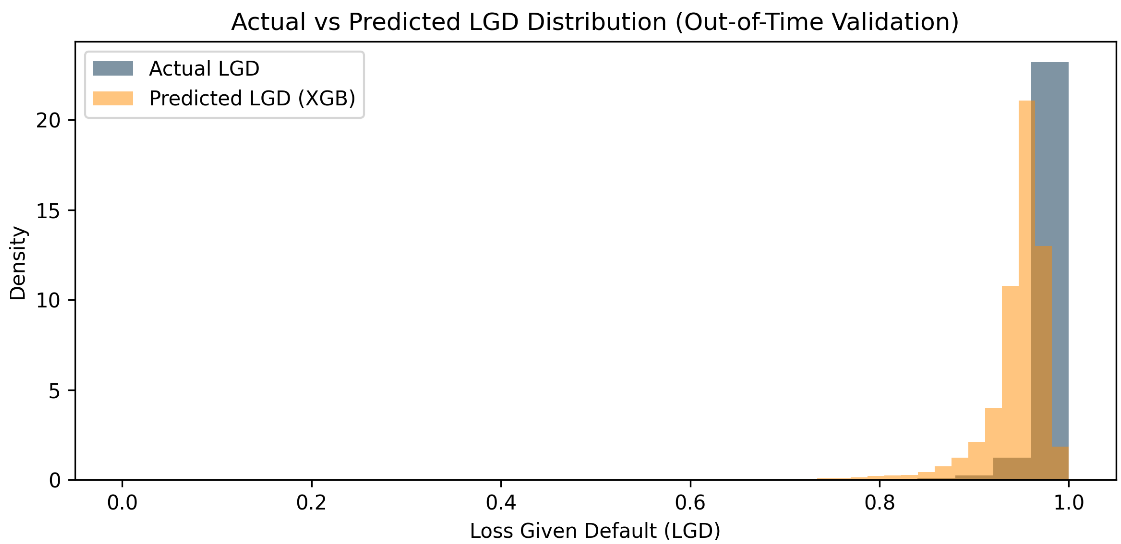
2. Model Performance Evaluation

Performance is compared across the primary XGBoost model and the benchmark Random Forest model on both In-Time (Train) and Out-of-Time (Validation) populations.

Performance Metric	XGBoost (Primary)	Random Forest (Benchmark)
Train RMSE	0.0992	0.1160
Train MAE	0.0500	0.0563
Train R^2 Score	0.4282	0.2178
Validation (OOT) RMSE	0.0728	0.0667
Validation (OOT) MAE	0.0542	0.0516
Validation (OOT) R^2 Score	-1.5835	-1.1662

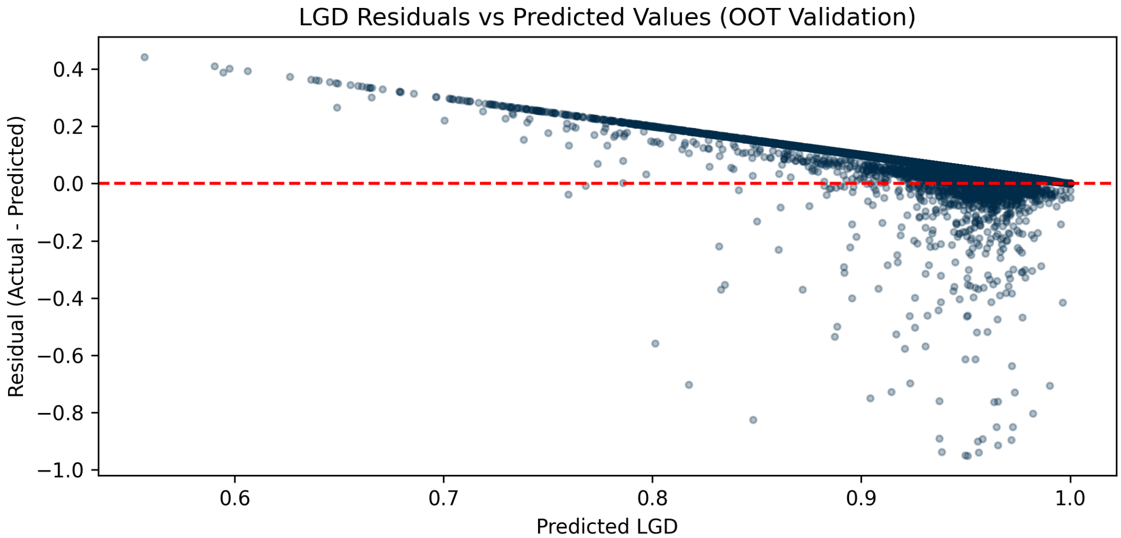
3. Model Diagnostics & Visualizations

The chart below displays the actual vs. predicted LGD distribution. Due to the high recovery skew in the portfolio (where many defaults recover little to nothing), the actual target distribution has a significant spike near 1.0. The XGBoost model represents a conservative expectation of loss severity.


Actual vs Predicted LGD Distribution (Out-of-Time Validation)
Actual LGD
Predicted LGD (XGB)
Density
0
5
10
15
20
0.0
0.2
0.4
0.6
0.8
1.0
Loss Given Default (LGD)

4. Residual Analysis & Diagnostic Checks

The residual plot displays the difference between actual LGD and predicted LGD. A uniform residual band indicates stable and unbiased predictions across the credit bands.



5. Model Governance & Limitations

In accordance with Basel and IFRS 9 guidelines, LGD models are calibrated using default cohorts only. Because underwriting is conducted at application time, the model only uses variables available before funding, avoiding any post-origination features that would cause data leakage in simulation workflows.